

BNNAction-Net: Binary Neural Network on Hands Gesture Recognitions

Federico Fontana
fontana.f@di.uniroma1.it
Sapienza, University of Rome
Italy, Rome

Alessandro Di Matteo*
alessandro.dimatteo1@graduate.univaq.it
University of L'Aquila
L'Aquila, Italy

Luigi Cinque
cinque@di.uniroma1.it
Sapienza, University of Rome
Rome, Italy

Giuseppe Placidi
giuseppe.placidi@univaq.it
University of L'Aquila
L'Aquila, Italy

Marco Raoul Marini
marini@di.uniroma1.it
Sapienza, University of Rome
Rome, Italy

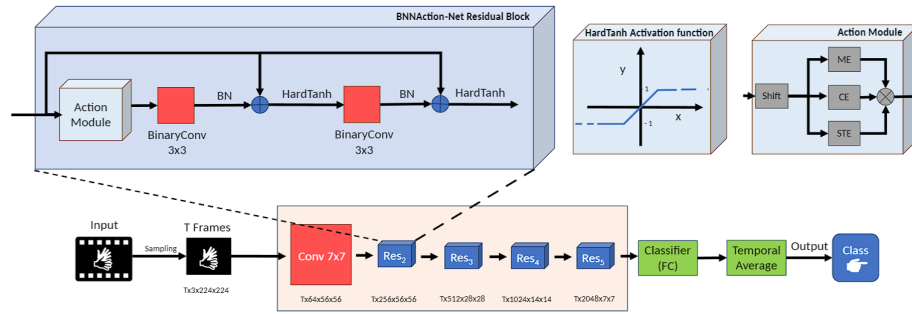


Figure 1: Pipeline and architecture of the proposed system.

ABSTRACT

With the rise of wearable technology and real-time gesture recognition, lightweight and efficient models are essential. Traditional approaches struggle with computational demands and power consumption. We present BNNAction-Net, a hand gesture recognition system using Binary Neural Networks (BNNs) to reduce computational complexity. Evaluated on the EgoGesture dataset, our system simulates a real use case with a headset and frontal RGB-D cameras. Optimized with binary layers, pooling, and normalization, it achieves accuracy comparable to floating-point networks with lower resource consumption. Our findings highlight the efficiency of BNNs for wearable devices without significant accuracy loss.

ACM Reference Format:

Federico Fontana, Alessandro Di Matteo*, Luigi Cinque, Giuseppe Placidi, and Marco Raoul Marini. 2024. BNNAction-Net: Binary Neural Network on Hands Gesture Recognitions. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Posters (SIGGRAPH Posters '24)*, July 27–August 01, 2024, Denver, CO, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3641234.3671047>

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
SIGGRAPH Posters '24, July 27–August 01, 2024, Denver, CO, USA
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-0516-8/24/07
<https://doi.org/10.1145/3641234.3671047>

*Corresponding author.

1 INTRODUCTION

Gesture recognition technology enables digital devices to interpret human gestures as input commands [Li et al. 2018], allowing seamless interaction with machines. Hand Gesture Recognition (HGR) is widely used in Virtual Reality (VR) for rehabilitation, gaming, remote control, and education applications. Integrating HGR into untethered VR headsets enhances user experience by providing autonomy and unrestricted hand movement control. However, these systems are limited by lower processing power. HGR systems must be optimized for the Computer Vision Platform to ensure reliability and accuracy. This study introduces BNNAction-Net, a novel ego-view HGR architecture using a Binary Neural Network (BNN) optimized for low-resource devices. BNNAction-Net efficiently processes gesture recognition data under limited resource conditions by exploiting BNNs [Rastegari et al. 2016] and action module [Wang et al. 2021].

2 METHOD

Our method innovates HGR by adapting the ActionNet [Wang et al. 2021] architecture tailored to the EgoGestures dataset [Zhang et al. 2018]. The EgoGesture dataset is a collection of 24161 hand gesture samples grouped into 83 different classes and captured from an egocentric perspective using an RGB-D Camera in indoor and outdoor environments. The key modification involves replacing the conventional backbone with a BNN, known for its efficiency. In detail, it consists of a binarized ResNet34 [Xu et al. 2021], enhancing the network for accurate HGR. Further amplification of our model's performance is achieved by applying Knowledge Distillation (KD) and leveraging a pre-trained ActionNet model based on ResNet50 as a teacher.

Table 1: Comparison with other state-of-the-art methods, where Ops represents the number of operations for each gesture recognition inference on a sample of size $8 \times 3 \times 224 \times 224$.

Method	Ops	Param	Acc %
TSN[Lin et al. 2019]	33G	23.68M	83.1
TSM[Jiang et al. 2019]	33G	23.68M	92.1
ACTION-Net[Wang et al. 2021]	34.7G	28.08M	94.2
BNNAction-Net(Our)	3.2G	18.69M	92.3
	−90.77%	−33.44%	−1.86

Table 2: Experiments investigating the effects of varying network components, loss functions, and optimizers on accuracy.

Variation	Ops	Param	Acc %
Baseline			
ActionNet with backbone	3.21G	18.62M	75.6
XNOR-Net on ResNet34 [Rastegari et al. 2016]			
+ Learnable Bias and weights clipping [Xu et al. 2021]	3.29G	18.69M	89.3
+ Knowledge Distillation	3.29G	18.69M	90.4
- Cross Entropy Loss	3.29G	18.69M	91.5
+ Hero Lion Optimizer [Chen et al. 2023]	3.29G	18.69M	92.3

2.1 Knowledge Distillation

KD transfers rich feature understanding from a complex ActionNet model, ResNet50 (teacher), to our modified BNNActionNet model (student), as described in [Hinton et al. 2015]. Based on ResNet50, the teacher model was trained on the EgoGestures dataset. The final loss is the Mean Squared Error loss calculated between the logits of the student’s network and those of the teacher’s network. Results in Table 2 show that the KD loss is more effective when used alone, eliminating the cross-entropy loss from our model.

2.2 Implementation Details

The entire model is implemented using PyTorch¹. Training is performed on the EgoGestures dataset, with the model being evaluated using accuracy. The model used as a backbone is the binarized ResNet34 [Xu et al. 2021] pre-trained on ImageNet. As illustrated in Fig. 1, this network design results in significant information loss. To mitigate this issue, we introduced additional skip connections to enhance the information flow. The optimizer used is Hero Lion [Chen et al. 2023] with a learning rate of 0.01, batch size of 64, and weight decay of 0.

2.3 Evaluation

Our model, BNNAction-Net, was rigorously evaluated using the EgoGestures dataset. We used the official test/train split. The outcomes, presented in Table 1, demonstrate a notable enhancement in HGR, confirming the efficacy of our architectural adjustments. These modifications enabled the network to operate with ten times fewer operations while incurring only a minimal decrease in accuracy. Additionally, Table 2 details the impact of each modification on the final accuracy.

3 CONCLUSION

We developed and evaluated an efficient system for HGR from an ego-view perspective using BNNs. Our findings show that BNNs significantly reduce computational demands with minimal accuracy loss. The system’s low power consumption and reduced processing needs make it ideal for standalone head-mounted displays (HMDs). Future work will refine the system’s accuracy while maintaining its efficiency and explore its integration with real

HMDs, such as using Huawei Bolt² for Android on Meta Quest. Finally, expanding the gesture library and improving user feedback mechanisms will enhance usability and user experience in practical applications.

ACKNOWLEDGMENTS

The research leading to these results has received funding from Project “Eco-sistema dell’innovazione - Rome Technopole” financed by EU in NextGenerationEU plan through MUR Decree n. 1051 23.06.2022 - CUP H33C22000420001.

REFERENCES

- Xiangning Chen, Chen Liang, Da Huang, Esteban Real, Kaiyuan Wang, Yao Liu, Hieu Pham, Xuanyi Dong, Thang Luong, Cho-Jui Hsieh, Yifeng Lu, and Quoc V. Le. 2023. Symbolic Discovery of Optimization Algorithms. *arXiv:2302.06675* [cs.LG]
- Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. 2015. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531* (2015).
- Boyuan Jiang, MengMeng Wang, Weihao Gan, Wei Wu, and Junjie Yan. 2019. Stm: Spatiotemporal and motion encoding for action recognition. In *Proceedings of the IEEE/CVF international conference on computer vision*. 2000–2009.
- Gongfa Li, Hao Wu, Guozhang Jiang, Shuang Xu, and Honghai Liu. 2018. Dynamic gesture recognition in the internet of things. *Ieee Access* 7 (2018), 23713–23724.
- Ji Lin, Chuang Gan, and Song Han. 2019. Tsm: Temporal shift module for efficient video understanding. In *Proceedings of the IEEE/CVF international conference on computer vision*. 7083–7093.
- Mohammad Rastegari, Vicente Ordonez, Joseph Redmon, and Ali Farhadi. 2016. Xnor-net: Imagenet classification using binary convolutional neural networks. In *Euro-pean conference on computer vision*. Springer, 525–542.
- Zhengwei Wang, Qi She, and Aljosa Smolic. 2021. Action-net: Multipath excitation for action recognition. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*. 13214–13223.
- Zihan Xu, Mingbao Lin, Jianzhuang Liu, Jie Chen, Ling Shao, Yue Gao, Yonghong Tian, and Rongrong Ji. 2021. Recu: Reviving the dead weights in binary neural networks. In *Proceedings of the IEEE/CVF international conference on computer vision*. 5198–5208.
- Yifan Zhang, Congqi Cao, Jian Cheng, and Hanqing Lu. 2018. Egogesture: a new dataset and benchmark for egocentric hand gesture recognition. *IEEE Transactions on Multimedia* 20, 5 (2018), 1038–1050.

¹<https://pytorch.org/>

²<https://github.com/huawei-noah/bolt>